

Safe Alone, Unsafe Together: Minimal Dynamic Incompatibility in SG-to-GFL Replacement Portfolios

TX4 research campaign

P10 — Safe alone, unsafe together: blind closure + contextual return

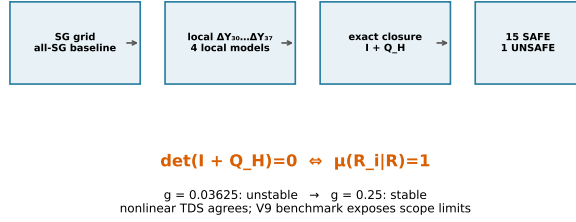


Fig. 1. Campaign logic: local replacements are composed through a common network closure, then checked against full-order dynamics.

Abstract—Replacement safety is not necessarily compositional. In a frozen IEEE-39 phasor-domain model, four local SG-to-GFL replacements form an unstable portfolio although each of its fifteen proper subsets is stable. We build one all-SG network kernel and four local replacement models, then predict the four-bus verdict before full-order reveal. The prediction is exact over all 16 portfolios. A classical contextual-return factorization identifies the collective closure responsible for the failure, and a nonlinear phasor-domain load-pulse test preserves the stable/unstable/stable ordering. A 512-portfolio benchmark is deliberately negative: accuracy is 77.15%, with 116 false-safe cases and no exact blocker-family recovery. The contribution is therefore a scoped model-specific mechanism and prediction package, not a universal portfolio predictor.

I. INDUSTRIAL QUESTION

As synchronous generation is replaced by grid-following converters, a plan that checks each replacement alone may miss a collective dynamic failure. The planning question is discrete: which finite set of local replacements is the first incompatible set, while load, dispatch, and network remain fixed?

This question is deliberately narrower than a generic grid-strength study. The object being tested is a finite replacement portfolio under one frozen policy, not an uncertainty distribution. The evidence is consequently reported as an exact finite census and as a preregistered blind test.

II. FROZEN MODEL AND WITNESS

The IEEE-39 phasor-domain DAE uses the frozen TX4 synchronous-machine, AVR, and ten-state GFL semantics. At P4, $(g, k, t, h) = (0.03625, 1.425, 1.5, 1)$, $H4=\{30, 33, 35, 37\}$

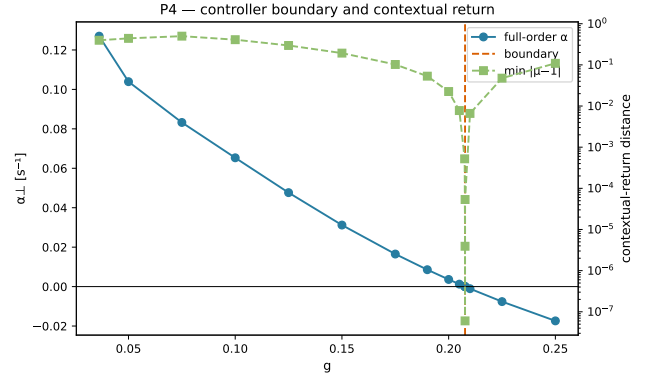


Fig. 2. Full-order spectral abscissa and contextual-return boundary under the one-coordinate g sweep.

has $\alpha = +0.127006 \text{ s}^{-1}$ at 0.622280 Hz. Every proper subset is stable; the four 7-of-8 predecessors have alpha values - 0.204688, -0.175508, -0.167856, and -0.203009 s^{-1} .

The four candidate buses are 30, 33, 35, and 37. The full-order test resolves the equilibrium before extracting the spectrum; the reduced test uses the same operating-point-matched local substitutions. This common setup matters because a portfolio verdict can otherwise be confounded with a different dispatch or initialization.

III. REDUCED CLOSURE

Let $T_0(s)$ be the all-SG network port and $K = E^T T_0^{-1} E$. Local substitutions give D_S and $M_S = D_S K_{SS}$. After diagonal normalization, $Q_S = (I + D_S K_d)^{-1} D_S K_o$. This is an exact determinant factorization under regularity assumptions; the algebra is classical.

The representation is useful because it reuses one network kernel and changes only local device terms. It is not a claim that a reduced determinant has the same spectral-abscissa ordering for every portfolio or operating point.

IV. BLIND PREDICTION

The firewall allowed only the model, baseline network, local devices, and port construction before reveal. The V4 predictor enumerated all 16 subsets: 15 stable predictions and one unstable H4 prediction. Full-order reveal gave 16/16, zero false-safe, zero false-unstable, exact H, and exact $\kappa=4$. The H4 reduced frequency was 0.622279671 Hz, agreeing with full order to 1.4×10^{-9} Hz.

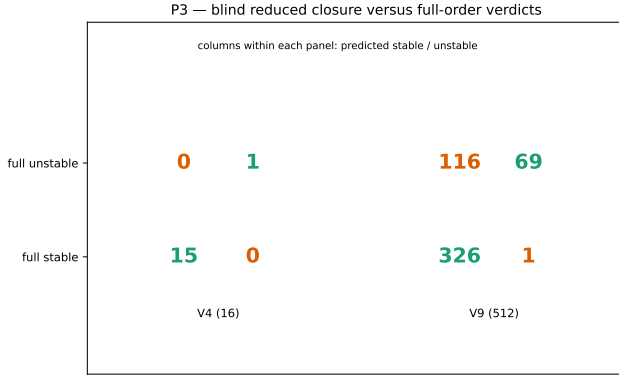


Fig. 3. Blind V4 reduced-closure verdicts versus full-order reveal.

The blind protocol was frozen before the reveal. It enumerated all 16 V4 subsets, wrote and hashed the prediction files, and only then loaded the full-order answer tables. The V4 result is therefore an exact finite pre-reveal prediction result; the later V9 run is a transfer stress test.

V. CONTEXTUAL MECHANISM

For $R = H \setminus \{i\}$, define $G_R = (I + Q_{RR})^{-1}$ and $\mathcal{R}_{i|R} = Q_{iR}G_RQ_{Ri}$. Then

$$\det(I + Q_H) = \det(I + Q_{RR}) \det(I - \mathcal{R}_{i|R}). \quad (1)$$

At the boundary, the collective factor has singular value 2.2×10^{-8} while local factors remain regular. The nearest contextual-return eigenvalue is -0.999999965 . The H4 controller boundary is $g = 0.207681405$ in full order; the reduced-only reproduction is 0.207681386 . Its run timing was post-reveal, so this agreement is reported with a procedural limitation.

At the boundary the local factor remains regular while the collective factor approaches singularity. This is the mechanism-level distinction: the failure is not attributed to a single isolated inverter pole. The factorization is standard block algebra, and its empirical role here is to expose the portfolio-conditioned return path.

VI. CONTROLLER AND NONLINEAR VALIDATION

Changing only g to 0.25 gives $\alpha = -0.017385 \text{ s}^{-1}$. Under the frozen 2% bus-20 active-load pulse for 0.2 s , the proper subset is stable, H4 is unstable, and retuned H4 is stable. Growth-sign agreement is 3/3. This is nonlinear phasor-domain TDS, not EMT validation.

The common disturbance is a two-percent active-load pulse at bus 20 for 0.2 s . The three declared cases are a proper triple, H4 at the frozen controller, and H4 after the fixed g retuning. The estimated growth-rate sign agrees with the linear alpha sign in all three cases. This supports the ordering for the tested disturbance; it does not establish transient security for arbitrary faults, pulses, or EMT representations.

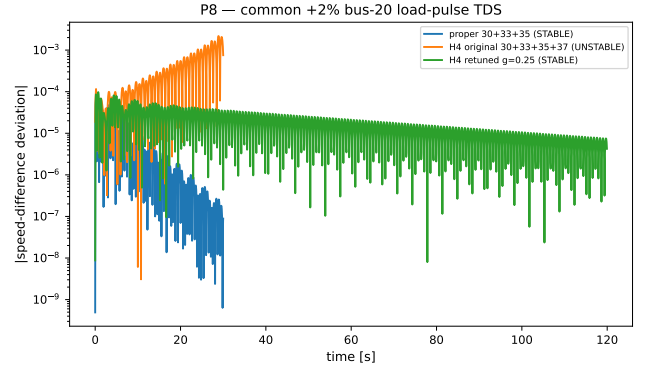


Fig. 4. Common-disturbance nonlinear phasor-domain TDS for a proper subset, H4, and retuned H4.

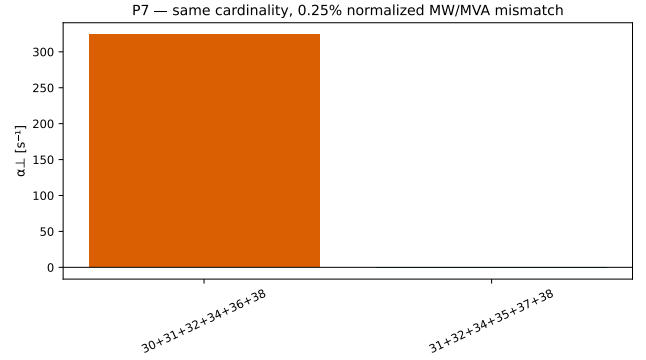


Fig. 5. Near-matched aggregate replacement pair with opposite full-order verdicts.

VII. AGGREGATE COUNTEREXAMPLE AND V9 LIMIT

The nearest fixed-rule V9 pair has six replacements and 0.2513% normalized MW/MVA mismatch: one portfolio is unstable ($\alpha = +324.249$) and the other stable ($\alpha = -0.1664$). There is no exact aggregate match. Over all V9 subsets, accuracy is 395/512, false-safe count 116, false-unstable count 1, and exact H recovery fails. Thus the predictive claim remains flagship-scoped.

The aggregate comparison reinforces the same caution. A fixed-rule, same-cardinality V9 pair differs by only 0.2513% in normalized MW/MVA distance, yet one member is stable and the other unstable. This is not an exact equal-penetration proof, because exact MW/MVA matching was unavailable; it is a near-match counterexample against reading penetration as a sufficient descriptor. Conversely, the V9 census has 116 false-safe cases, so the reduced closure should not be marketed as an exact scalable classifier.

VIII. DISCUSSION AND LIMITATIONS

The critical mode is an inter-area electromechanical mode with network-mediated converter participation; numerical residual and eigensolver checks do not indicate an index pathology. The model is one IEEE-39 phasor-domain implementation, ANDES support is limited, EMT remains unresolved, and the reduced method is slower than full order. The

value is diagnostic: it shows what fails, why the collective closure fails, and how a fixed controller coordinate moves the boundary.

IX. CONCLUSION

The result supports the scoped statement: replacement safety is a portfolio property in the frozen TX4 model, and a local-device/network closure can predict the minimal four-bus failure before full simulation. Broader model and network validation is required before a journal-level generalization.

REFERENCES

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